

Implementation of Agile Software Development in the Design of Management Information Systems Inventory and Finance at P2MKP Alang-Alang Tumbuh Subur [†]

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[†] Presented at the 8th Mechanical Engineering, Science and Technology International Conference, Padang Besar, Perlis, Malaysia, 11–12 December 2024.

Abstract: P2MKP Alang-alang Tumbuh Subur is an MSME that produces various types of food made from catfish. Inventory recording in these MSMEs is still carried out using simple Excel, which causes various problems, such as differences in real stock and recording stock, longer recording time, and vulnerable data loss. Therefore, it is necessary to improve the recording system. This research aims to design a management information system that is informative and easily accessible, so that inventory and financial records in these MSMEs can be performed quickly and flexibly. The method used is the agile software development method, which prioritizes communication with related parties with several sprints. This method ensures that the system can be developed quickly, precisely, and in accordance with the wishes of the user. The result of this research is a system that contains product stock management pages, employee data, equipment, and business financial transactions. With CRUD (create, read, update, delete) and report printing features, users can easily change and print reports. In addition, there is an access code feature so that data security is guaranteed. With this system, it is expected that MSMEs can increase the efficiency and accuracy of inventory and financial records and support the smooth operation of the company as a whole.

Keywords: management information system; inventory and finance; agile software development



Academic Editors: Noor Hanita Abdul Majid, Agus Dwi Anggono, Waluyo Adi Siswanto, Tri Widodo Besar Riyadi, Mohammad Sukri Mustapa, Nur Rahmawati Syamsiyah and Afif Faishal

Published: 18 February 2025

Citation: Ramadhani, D.S.; Aulia, N.; Putri, A.S.; Nugroho, M.T.; Sakti, A.S.A.; Satriyono, R.D.A.P. Implementation of Agile Software Development in the Design of Management Information Systems Inventory and Finance at P2MKP Alang-Alang Tumbuh Subur. *Eng. Proc.* **2025**, *84*, 62. <https://doi.org/10.3390/engproc2025084062>

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1. Introduction

The development of technology today has grown rapidly, affecting all aspects of human life. Information technology plays a very important role in human activities so that all factors of life cannot be separated from technology. Technology has various functions, such as recording information, processing information, generating information, and storing information [1]. One of the rapidly growing technologies is internet information media. The internet is a network that functions to connect one electronic media with another so that the data transfer process can be carried out precisely and quickly with a certain frequency. Information technology and the internet are an epic combination in creating a quality system that is more effective and efficient. With information technology, it can improve company performance in processing all information so that it can increase company competitiveness [2].

Administration is one of the fields that requires sophisticated information technology. In daily activities, administration requires technology to record or store all forms of data in the company more effectively and efficiently. With the existence of qualified information technology, it is hoped that it can help agencies or companies in completing several tasks related to services to their users [3]. One of the tasks carried out by the administration department in an MSME or company is recording inventory and reports on business income and expenses. Inventory management is an activity carried out by a company or business actor to meet the material needs of company activities for both the production and sales processes so that they can be maximally fulfilled. Inventory management is also useful for decision making on every problem in the company by looking at the condition of the company [4].

P2MKP Alang-alang Tumbuh Subur is an MSME engaged in food, especially catfish processing, which has various products with catfish as the basic ingredient, such as shredded catfish, catfish chips, fish nuggets, and frozen fish fillets. Every administrative activity in MSMEs such as the receipt and expenditure of raw materials, the employee payroll process, and the process of selling or buying products must be documented carefully to facilitate the smooth running of administrative work. So far, Alang-alang Tumbuh Subur MSMEs have used Microsoft Excel to record all the administration in MSMEs and there are even some records that are performed manually. Microsoft Excel is software that is used to record or process data and can analyze company data. But in reality, recording inventory using Excel has shortcomings that come from several factors, both from the application and human error itself.

P2MKP Alang-alang Tumbuh Subur only has one admin staff, which causes the admin workload to be very heavy. This admin performs a variety of tasks such as recording the income and expenditure of raw materials and finished products, assisting the production process, and also receiving customer orders. The imbalance between the number of workers and the available labor causes administrative management to be not optimal and the risk of errors increases. This will have an impact on MSMEs that experience losses because the operational process does not run smoothly. In addition, heavy workloads on workers can cause fatigue and stress, which can ultimately affect the quality of performance and employee welfare. This situation can also make it difficult for admins when there are urgent situations or sudden changes, because no one can back up the work of the admin. As a result, MSMEs can face delays in order handling, errors in data recording, and potential loss of customers due to inadequate service.

The use of Microsoft Excel in managing the administration of MSMEs is still relatively new so it is necessary to reassess this recording system. In addition, the number of workers who understand and are proficient in using Excel is still very limited. Ignorance of Excel features and lack of training on the use of Excel make recording not optimal. This results in slower and error-prone administrative processes that can ultimately have a negative impact on the performance of MSMEs. Therefore, additional time and resources are needed to learn to master Excel proficiently, which could be used for other operational activities.

The problem faced by admin staff while using Excel in the recording system is the inefficiency of the recording system because the features in Excel are not many and do not support all recording activities in MSMEs. One example of inefficient inventory recording is having to open item codes one by one in Microsoft Excel. Each product produced in MSMEs has its own code to distinguish between one product and another. The many types of products produced by MSMEs make it difficult for admin staff to record product inventory because they have to see the item code and its name one by one in the product list notes and it is difficult to memorize all the existing product codes. In addition, data stored on the computer are at risk of being lost whether done intentionally or not. Accidental

deletion of data files by admin staff or computer errors can eliminate item records so that the final information on the inventory does not match what is in the warehouse [5]. This is certainly detrimental to MSMEs because these data are important assets for the company.

Another problem at P2MKP Alang-Alang Tumbuh Subur is that there is no detailed recording of business income and expenditure reports. Currently, there is only a record of outgoing goods/products and there is no detailed data collection of how much income is obtained and what expenses have been made by business actors. Whereas financial information is very useful to know the state and development of the business, income and expenditure reports are very useful for companies to make decisions in every problem [6]. So, if these finances are not recorded properly and securely, the company will have difficulty in identifying areas that require improvement or cost savings. Without clear financial records, MSMEs cannot assess overall financial performance, which can lead to inappropriate and unsubstantiated business decisions.

Based on the problems that have been described, this research aims to design a website-based inventory and financial management information system using the PHP program language. With this information system, it is hoped that it can help admin work in recording MSME inventory and managing cash flows in and out of MSMEs more effectively and efficiently than the current system. The formation of an information system that is integrated with the programming language is expected to ensure that data processing can run more smoothly, where the level of accuracy is higher so as to increase the level of accuracy of the data produced. Data will be stored centrally in a database so that data searches will be faster and data security is more guaranteed with sophisticated data security technology. Data management that has not been optimized needs to be improved and updated through increased functionality in a new system that is integrated with current technology [7]. So, this research is expected to build an information system that can meet the needs of business actors functionally or non-functionally and produce software applications that are tested for feasibility [8].

This research was conducted using the System Development Life Cycle (SDLC) method through the agile software development approach. The SDLC is a general method that is often used in the development of information systems. The SDLC consists of four stages, namely planning, analysis, design, and implementation. Meanwhile, agile software development is an approach in the SDLC to create a software system that is a development of the traditional waterfall method. This method makes it easier for researchers to design programs because it involves business actors actively in their development so that it will minimize functionality errors in the program [9]. The advantage of the agile method is that it is more flexible to changes in the manufacturing process until the system is in accordance with customer wishes, besides that the time required to make the system is faster and more efficient because it is performed in iterations. This method can also save costs because the development process is carried out gradually and repeatedly. Meanwhile, the shortcomings in the agile method are the difficulty of monitoring and controlling the development process due to frequent changes. In addition, the documentation of the development process is incomplete so that the documented information is less detailed [10]. By using this method, it is hoped that software development can be carried out more systematically, be responsive to changes, and the results are in accordance with user desires [11]. Based on the description of the problems above, the agile software development method is used to conduct research on designing information systems to improve data processing on inventory records and reports on income and expenses at P2MKP Alang-alang Tumbuh Subur.

2. Research Method

This research was conducted at P2MKP Alang-alang Tumbuh Subur, which is located at Randusari, Teras, Boyolali Regency, Central Java. This research focuses on the administrative part of MSME management, which is tasked with managing all important data on MSMEs such as incoming goods data, outgoing goods data, income data, and expenditure data every day. The sources in this research are the owner and MSME admin staff. The method used is the agile software development method.

2.1. System Requirements Analysis

System requirements analysis is a critical process in software development that aims to identify and define the needs and expectations of users and other stakeholders [12]. At this stage, the researcher analyzes what functional and non-functional needs the new system must meet. This analysis also involves determining the required software and hardware specifications, as well as identifying key features and functionality that the system must have to support business and operational goals.

2.2. System Design

This system design aims to design in more detail the components that are in the system. The system design stage is made using several tools.

2.2.1. Context Diagram

A context diagram is a diagram that describes the system as a whole. In designing a system, it is important to understand how the system works as a whole. Therefore, the first diagram designed in this study is a context diagram, to make it easier for researchers to understand the scope of the system to be created [13].

2.2.2. Data Flow Diagram

A Data Flow Diagram is a diagram to model the flow of data in the system. Researchers will design a DFD to document business processes and information flow in the company. The purpose of designing this system is to make it easier for researchers and other stakeholders to understand the workflow of the system to be designed [14].

2.2.3. Entity Relationship Diagram

An Entity Relationship Diagram serves to provide an overview of the components that will be stored in the database. An ERD has an important role in system development because it guides researchers in designing the system. By using an ERD, researchers can determine the entities, attributes, and relationships between entities needed to meet system needs [15].

2.2.4. Use Case Diagram

Use case diagrams are useful for describing the relationship or interaction between users and systems or programs. By using this diagram, researchers can identify actions or functions performed by system users or external entities [16]. In this context, the program is aimed at admin staff, so the actors in this program are admins and MSME owners. Researchers will map the functions of each actor.

2.2.5. CRUD Matrix

A CRUD Matrix is a table that shows the relationship between business processes and data entities used in a system. This table has four main components, namely create, read, update, and delete. The function of the CRUD Matrix is to make it easier for users to under-

stand the operations that can be performed on certain data entities, whether they include creating, reading, updating, or deleting data or only limited to certain operations [17].

2.2.6. Application Design Using Figma

Application design using Figma serves as a prototype before the system is created using the actual program [2]. Using Figma will help researchers to test the use of the system that will be discussed with MSMEs so that if there are designs or functions that are not in accordance with the needs of MSMEs, they can be revised quickly before making the real system.

2.3. System Development

Information system development is carried out using the Hypertext Preprocessor (PHP) programming language and MySQL database. The program is developed based on a dynamic website. This will make it easier for users because they do not need to understand programming syntax to change the information contained in the program. The system development will be assisted by using CodeIgniter and Bootstrap frameworks. System development will be carried out in stages, starting from the login menu, product stock menu, and other menus. MSMEs will play an active role during the web development process to ensure that no errors occur and the website developed is in accordance with the wishes of the user. The involvement of MSMEs in every stage of development is very important to ensure that all the features and functionality needed can be implemented appropriately.

2.4. System Testing

Program testing aims to detect the program whether an error occurs or not so that it can be identified and corrected early on. System testing in this study uses the BlackBox Testing method. The purpose of using this method is to evaluate the system based on the input and output produced and ensure that the system functions in accordance with the predetermined functional specifications [18].

3. Results and Discussion

3.1. System Requirements Analysis

At Alang-alang Tumbuh Subur MSME, the current inventory recording system uses the Microsoft Excel application, which utilizes the VLOOKUP formula. Meanwhile, the financial recording system is still manual with the use of a notebook held by the owner. Checking product stock is performed every day by workers. This manual system has disadvantages including being prone to errors if the data recording process is not careful and poor data structure that causes errors in data matching. In addition, the use of Excel has limited accessibility because it can only be opened via desktop and is prone to data security issues. The information needed by the system to be developed includes the following:

- a. Information about the recording concept used previously.
- b. Information about the concept of the system to be created.
- c. Information about what elements need to be displayed on the system.
- d. MSME data that are relevant and need to be included in the system.

The development of this inventory and financial recording system aims to make it easier for MSMEs to document data accurately and quickly. Some of the capabilities that this system must have include the following:

- a. The system is able to shorten the time for recording inventory and MSME transactions.
- b. The system is able to process data and present them as valid and accurate information.
- c. The system interface is user-friendly so that it is easy for users to operate.

- d. The system is able to store a lot of data safely and efficiently.
- e. The system can generate reports every time.

The software needed in this research are XAMPP version 8.1.4, Visual Studio Code version 1.92, Google Chrome 109, MySQL 8.0, and Figma 3.5.

3.2. System Design

3.2.1. Context Diagram

The context diagram describes the outline of the data flow in the system. In this system, there are two entities, namely the owner and admin. The context diagram for this management information system can be seen in the following Figure 1.

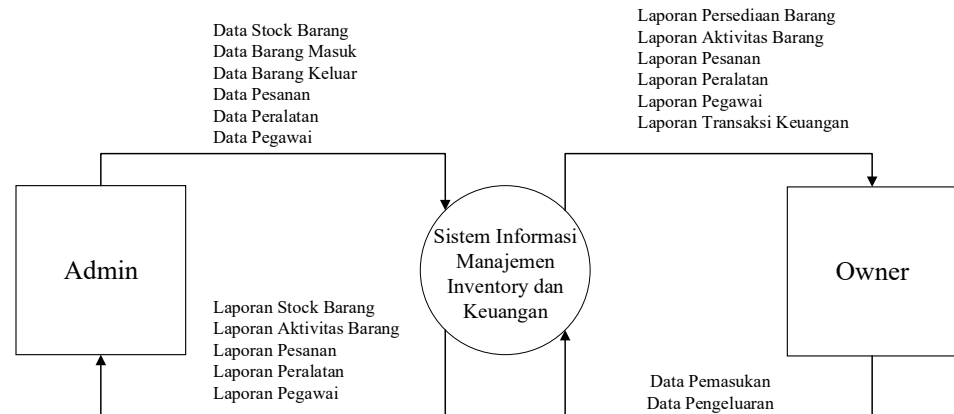


Figure 1. Context diagram.

3.2.2. Data Flow Diagram

A Data Flow Diagram is a diagram that is an elaboration of the context diagram and describes each component of a system and data flow including origin, destination, and storage. In this system, there are three core activities that can be seen in the following DFD (see Figure 2).

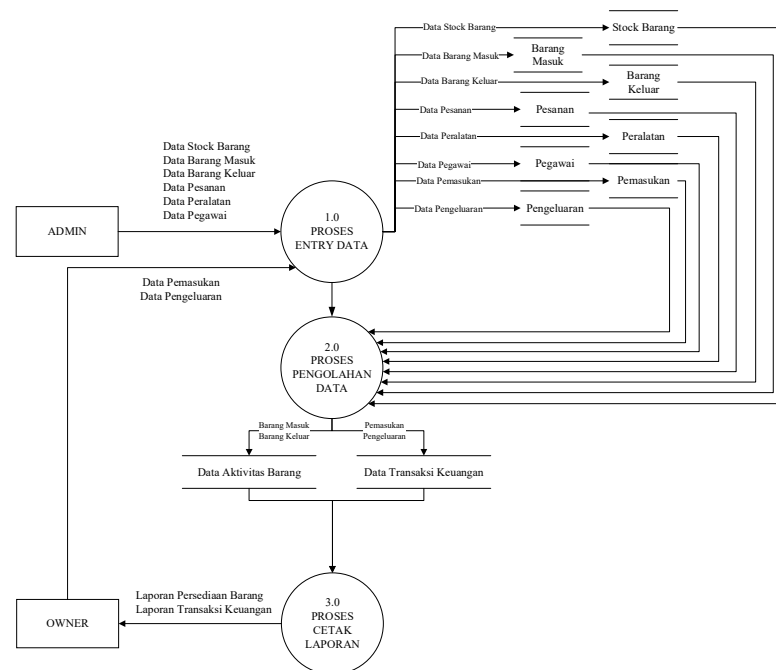


Figure 2. Data Flow Diagram.

3.2.3. Entity Relationship Diagram (ERD)

Based on the system requirements that have been analyzed by researchers, several tables are needed to store data. The relationship of the database tables in this system can be seen in the ERD below (see Figure 3).

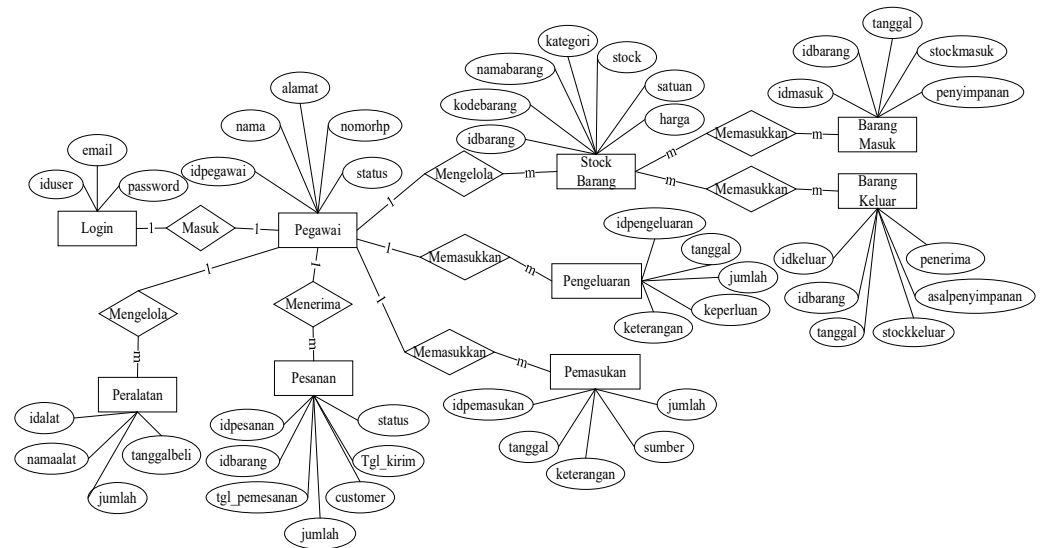


Figure 3. Entity Relationship Diagram.

3.2.4. Use Case Diagram

A use case diagram is a diagram that describes the interaction between actors and use cases in a system. Use case diagrams on the financial inventory management information system for UMKM Alang-alang Tumbuh Subur can be seen in the following figure (see Figure 4).

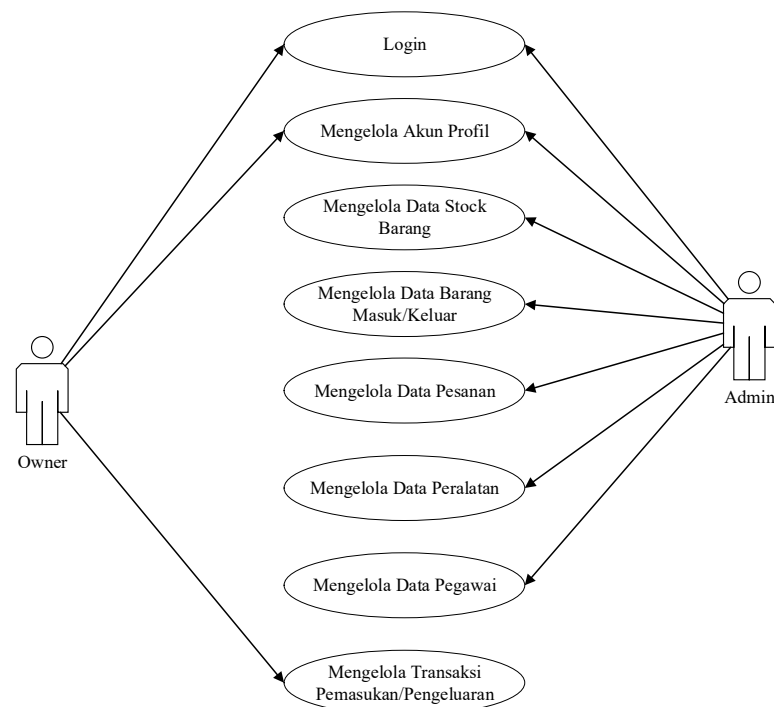


Figure 4. Use case diagram.

3.2.5. CRUD Matrix

A CRUD Matrix is a table that describes how the system process uses data in the system and provides important information on how data can be used by each entity.

3.3. System Development

3.3.1. Login Page

The login page is the first page displayed on the website if the user accesses the website. On this page, there are username and password fields to enter the account that has been registered (see Figure 5).

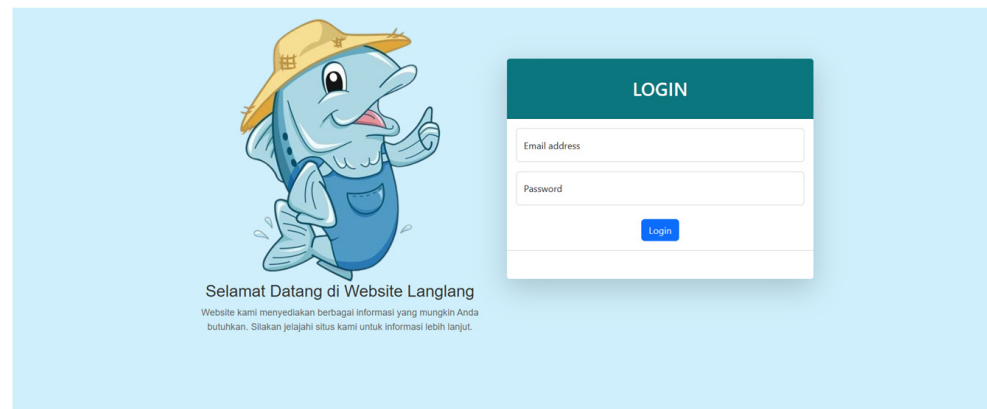


Figure 5. Login page.

3.3.2. Stock Page

The stock page is a page that displays a table that contains a list of stock products available at MSME Alang-Alang Tumbuh Subur (see Figure 6).

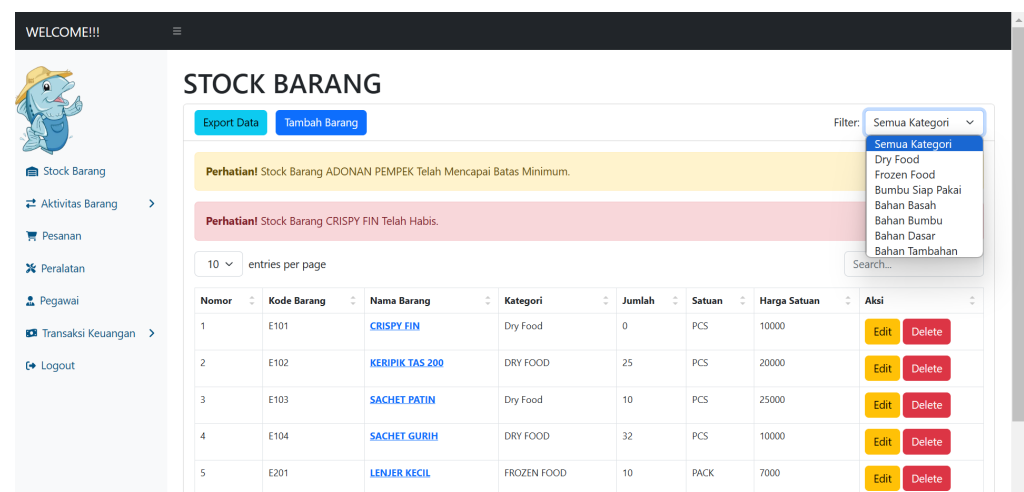


Figure 6. Stock page.

3.3.3. Item Detail Page

The item detail page is a page that provides a complete summary of a particular item that contains detailed information about the activities related to the item (see Figure 7).

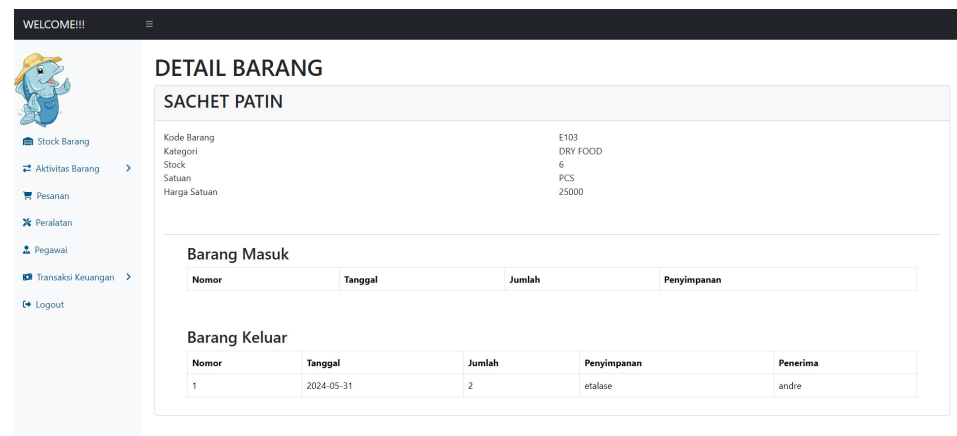


Figure 7. Item detail page.

3.3.4. Item Entry Page

The item entry page is a page that contains a list of items that have just entered the inventory (see Figure 8).

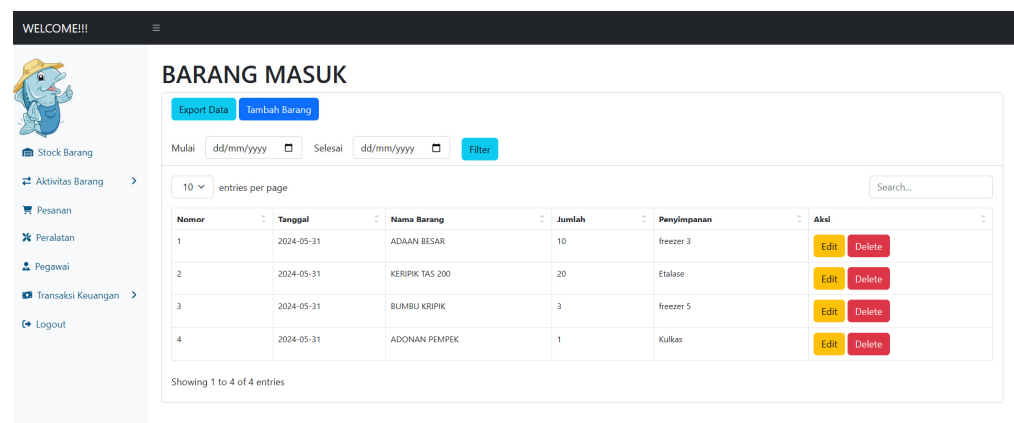


Figure 8. Item entry page.

3.3.5. Item Exit Page

The item exit page is a page that presents a list of items that have left the inventory (see Figure 9).

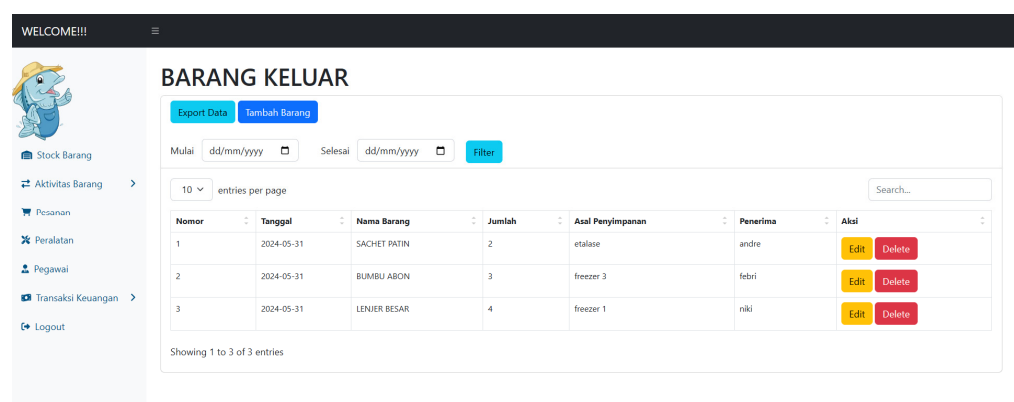


Figure 9. Item exit page.

3.3.6. Item Activity Report

The item activity report page contains a record of all item activity, both incoming and outgoing items. This page allows the owner to monitor the overall activity of goods and

review the goods activity report on a monthly basis or within a certain period of time (see Figure 10).

LAPORAN AKTIVITAS BARANG PERUSAHAAN

Start Date: End Date:

Search:

Nomor	Tanggal	Nama Barang	Penyimpanan	Jumlah Masuk	Jumlah Keluar
1	2024-05-31	ADAAN BESAR	freezer 3	10	
2	2024-05-31	KERIPIK TAS 200	Etalase	20	
3	2024-05-31	BUMBU KRIPIK	freezer 5	3	
4	2024-05-31	ADONAN PEMPEK	Kulkas	1	
5	2024-05-31	SACHET PATIN	etalase		2
6	2024-05-31	BUMBU ABON	freezer 3		3
7	2024-05-31	LENJER BESAR	freezer 1		4

Showing 1 to 7 of 7 entries Previous 1 Next

Figure 10. Item activity report.

3.3.7. Order Page

The order page is the page where orders from customers are recorded (see Figure 11).

WELCOME!!!

PESANAN BARANG

Pesanan On Process : 1

10 entries per page Search...

Nomor	Tanggal Pesan	Nama Barang	Jumlah	Pemesan	Tanggal Kirim	Status	Aksi
1	2024-07-02	LENJER BESAR	13	aris	2024-07-02	Berhasil	
2	2024-07-02	LENJER BESAR	12	aisyah	2024-07-10	ON_PROCESS	Selesai

Showing 1 to 2 of 2 entries

Figure 11. Order page.

3.3.8. Tool Page

The tool page is a page that lists the equipment owned by Alang-alang Tumbuh Subur MSMEs (see Figure 12).

WELCOME!!!

DATA PERALATAN

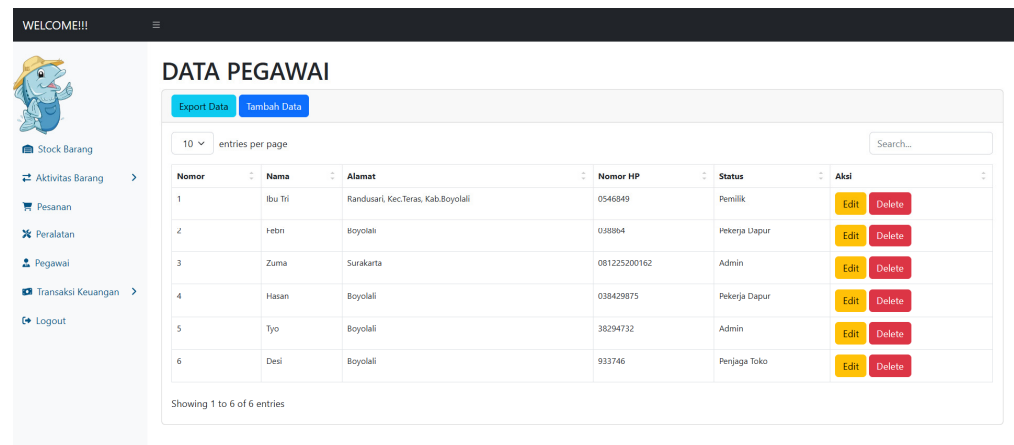
10 entries per page Search...

Nomor	Kode Barang	Nama Barang	Jumlah	Tanggal Beli	Aksi
1	A101	Maja	4	2024-05-01	Edit Delete
2	A102	Kompor Besar	3	2024-05-01	Edit Delete
3	A103	Set Pisau	5	2024-05-01	Edit Delete
4	A104	Loyang Besar	9	2024-05-01	Edit Delete
5	A105	Loyang Kecil	7	2024-05-01	Edit Delete
6	A106	Dandang Kukus	4	2024-05-01	Edit Delete

Figure 12. Tool page.

3.3.9. Staff Page

The staff page is a page that contains information about employee data that work at the Alang-alang Tumbuh Subur MSME (see Figure 13).



WELCOME!!!

DATA PEGAWAI

Export Data | Tambah Data

10 entries per page

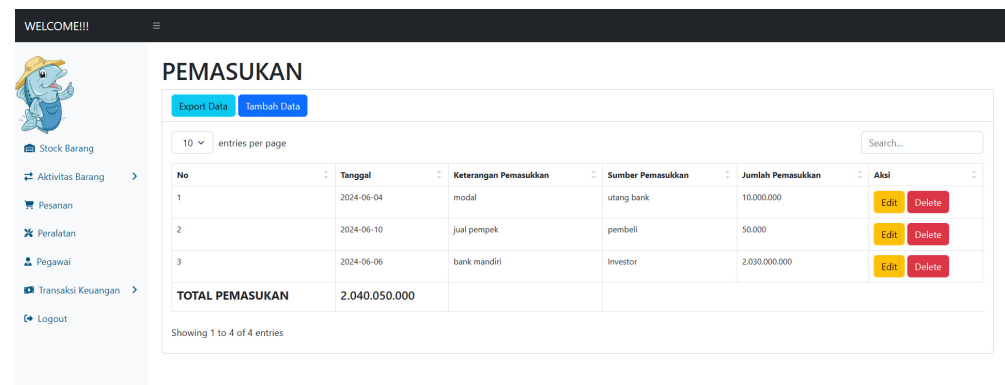
Nomor	Nama	Alamat	Nomor HP	Status	Aksi
1	Ibu Titi	Randusari, Kec.Teras, Kab.Bojolali	0546849	Pemilik	Edit Delete
2	Hebin	Bojolali	088864	Pekerja Uapuri	Edit Delete
3	Zuma	Surakarta	081225200162	Admin	Edit Delete
4	Hasan	Bojolali	038429875	Pekerja Dapur	Edit Delete
5	Tyo	Bojolali	38294732	Admin	Edit Delete
6	Desi	Bojolali	933746	Penjaga Toko	Edit Delete

Showing 1 to 6 of 6 entries

Figure 13. Staff page.

3.3.10. Income Page

The income page is a page that contains data on money coming into the company. The function of this page is to record all company revenue transaction activities (see Figure 14).



WELCOME!!!

PEMASUKAN

Export Data | Tambah Data

10 entries per page

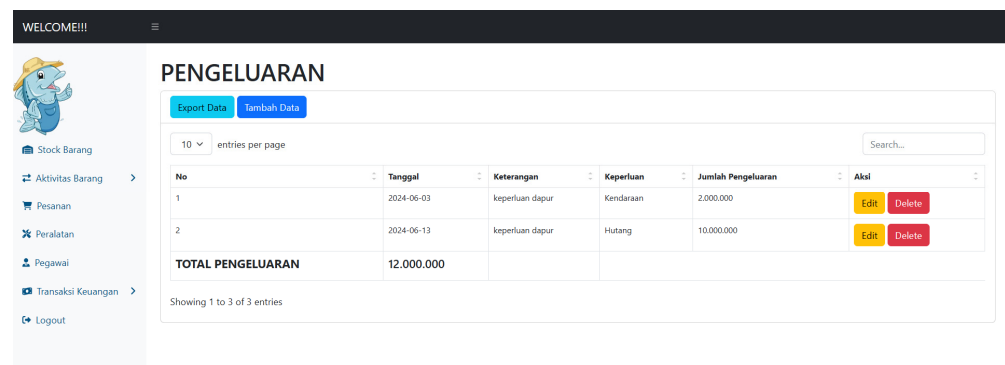
No	Tanggal	Keterangan Pemasukan	Sumber Pemasukan	Jumlah Pemasukan	Aksi
1	2024-06-04	modal	utang bank	10.000.000	Edit Delete
2	2024-06-10	jual pempek	pembeli	50.000	Edit Delete
3	2024-06-06	bank mandiri	Investor	2.030.000.000	Edit Delete
TOTAL PEMASUKAN		2.040.050.000			

Showing 1 to 4 of 4 entries

Figure 14. Income page.

3.3.11. Outcome Page

The outcome page is an important section that records all outgoing financial transactions, including payments to suppliers, employee salaries, and other operational costs (see Figure 15).



WELCOME!!!

PENGELUARAN

Export Data | Tambah Data

10 entries per page

No	Tanggal	Keterangan	Keperluan	Jumlah Pengeluaran	Aksi
1	2024-06-03	keperluan dapur	Kendaraan	2.000.000	Edit Delete
2	2024-06-13	keperluan dapur	Hutang	10.000.000	Edit Delete
TOTAL PENGELUARAN		12.000.000			

Showing 1 to 3 of 3 entries

Figure 15. Outcome page.

3.3.12. Financial Report

The financial report page is an important component that presents a comprehensive picture of the company's financial condition. With this page, owners can monitor the company's financial records, identify financial trends, and assess overall business performance (see Figure 16).

LAPORAN KEUANGAN PERUSAHAAN

Start Date: End Date: [Filter](#)

[Excel](#) [PDF](#) [Print](#) Search:

Nomor	Tanggal	Keterangan	Pemasukan	Pengeluaran
1	2024-06-03	keperluan dapur		2,000,000
2	2024-06-04	modal	10,000,000	
3	2024-06-06	bank mandiri	2,030,000,000	
4	2024-06-10	jual pempek	50,000	
5	2024-06-13	keperluan dapur		10,000,000

Showing 1 to 5 of 5 entries [Previous](#) [1](#) [Next](#)

[Back](#)

Figure 16. Financial reportpage.

3.4. System Testing

System testing is the stage where the website or application that has been created is tested before use to find out whether there are errors or deficiencies on the website [19]. System testing is a type of software testing that evaluates the overall functionality and performance of a complete and fully integrated software solution [20]. It tests if the system meets the specified requirements and if it is suitable for delivery to the end users [21]. This type of testing is performed after the integration testing and before the acceptance testing [22]. After testing the use of the system in actual conditions, there are significant differences between before and after using the system as described below [23].

4. Conclusions

Based on the results of the research and discussion related to the development of inventory and financial management information systems at P2MKP Alang-alang Tumbuh Subur (see Table 1), it can be concluded that the management information system developed is an informative system that supports all company operational activities so as to enable efficient data management. This system is equipped with create, read, update, and delete (CRUD) features that function properly, making it easier for the data recording and management process. In addition, data confidentiality is guaranteed by the existence of a website access code and a secret code to enter the financial page. With online access, MSME managers can open and check data anytime and anywhere, thus increasing flexibility and efficiency in business management. Because this system is general, it can be applied in various business sectors, such as in the culinary, craft, fashion, and service sectors that require effective and efficient inventory and financial management. This system still uses free hosting, so it is recommended for MSME owners to update to paid hosting for better website performance. This MSME does not yet have a patent brand, so it can be a theme for further research on marketing and copyright determination to assist MSMEs in determining effective marketing strategies and preventing plagiarism by irresponsible parties.

Table 1. Comparisons between before and after using the system.

	Before	After
Risk level	High risk of errors and data loss	Low risk of errors and data loss
Time	Inefficient	Efficient
Security level	No security system	Security is more guaranteed because access codes must be entered before entering the website
Accessibility	Data is difficult to access because it must be accessed through a PC	Data is easily accessible via online

Author Contributions: Conceptualization, D.S.R. and A.S.P.; methodology, N.A.; software, N.A.; validation, M.T.N., A.S.A.S. and R.D.A.P.S.; formal analysis, D.S.R.; investigation, R.D.A.P.S.; resources, A.S.A.S.; data curation, A.S.P.; writing—original draft preparation, D.S.R.; writing—review and editing, N.A.; visualization, N.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Acknowledgments: The author sincerely expresses gratitude to Universitas Muhammadiyah Surakarta for providing financial support that made this research possible. This funding has significantly contributed to the successful completion of the study, enabling the exploration and analysis of the subject matter in depth. The author appreciates the trust and resources extended by the university, which have been instrumental in achieving the research objectives.

Conflicts of Interest: The authors declare no conflict of interest.

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